6 Figure 6.1 shows a circuit that produces rectification of an alternating input voltage.

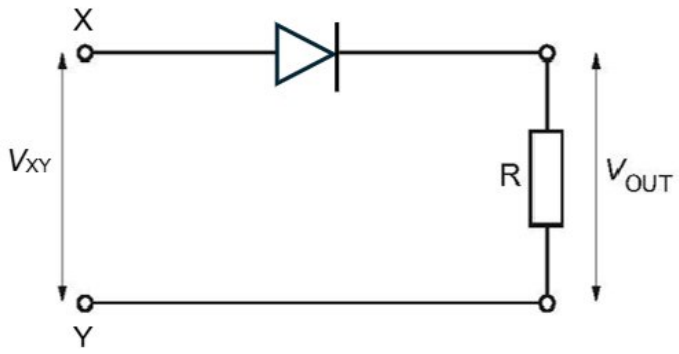


Figure 6.1

(a) The input voltage V_{XY} is the potential of X with respect to Y and is sinusoidal. V_{OUT} is the rectified voltage across resistor R.

The variation of V_{XY} with time t has amplitude V_0 and period T, as shown in Figure 6.2.

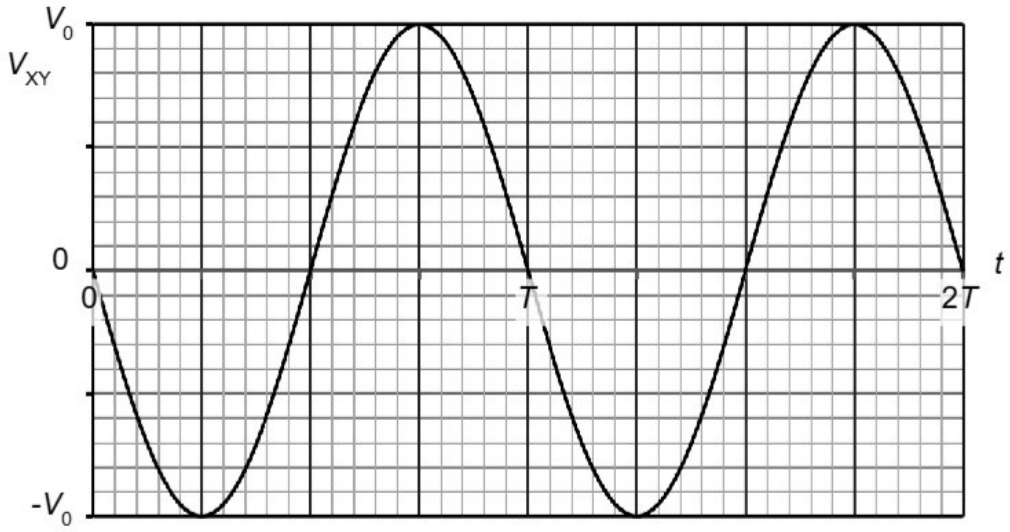


Figure 6.2

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(i) On Figure 6.3, sketch the variation with time t of the power P in the resistor between $t = 0$ and $t = 2T$.

Label the peak power P_0 in the resistor on your sketched graph.

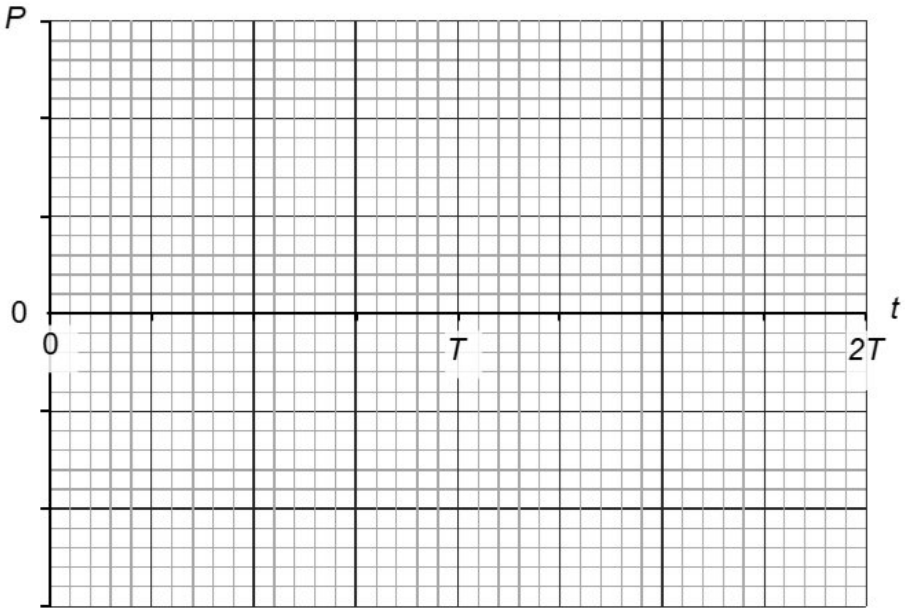


Figure 6.3

[2]

(ii) Use the answer in **(a)(i)** to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

.....

.....

..... [2]

(iii) The root-mean-square (r.m.s.) value of V_{XY} is 12 V.

Use the information in **(a)(ii)** to show that the r.m.s. value of V_{OUT} is 8.5 V.

[2]

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(b) To smooth the output voltage, a capacitor is connected in parallel to the resistor in Figure 6.1.

During each discharge cycle, the minimum value of the smoothed output voltage is 0.60 V_0,

(i) Figure 6.4 shows a duplicate of the graph of V_{XY} with time t.

On Figure 6.4, sketch the variation with time t of the smoothed rectified output voltage V_R across the resistor R between $t = 0$ and $t = 2T$.

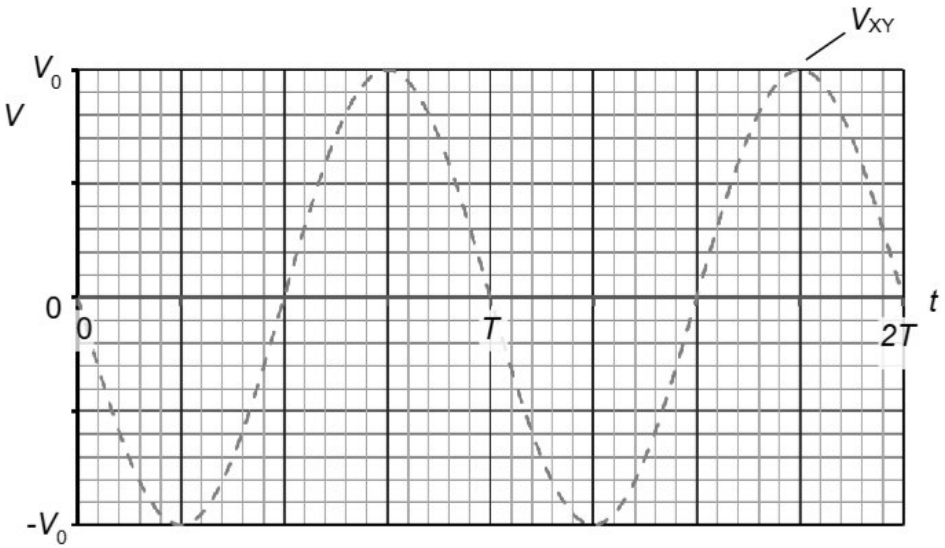


Figure 6.4

[1]

(ii) Determine the time constant of the smoothing circuit in terms of the period T.

time constant = ... [2]

[Total: 9]

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